SKILL 3

Feature Engineering and Exploratory Data Analysis

One Piece Bounty Dataset • Feature Creation • EDA • Modelling Notes

Input	165 rows and 3 raw columns
Engineered output	165 rows and 23 columns
Tier counts	12 Calamity • 34 Menace • 99 Wanted • 20 Unknown
Inequality	Gini coefficient 0.75; top 5 hold 42.2% of known bounty

1. Aim

Create analysis-ready features from the One Piece bounty dataset and explore distributions, bounty tiers, name-based signals, feature correlations and bounty inequality.

2. Feature groups

• Scaling: Bounty_Billions, Bounty_Log10, Bounty_Percentile, Bounty_Log_Zscore and Bounty_Log_MinMax.

• Binning: Tier, Tier_Code and one-hot tier indicators.

• Text: Name_Length, Word_Count, Has_D_Clan, Has_Parentheses and Is_Anonymous.

• Grouping: Name_Group, Group_Size and Is_Charlotte.

• Target: High_Bounty, based on the median log bounty.

3. Observation

The engineered dataset contains 165 records and 23 columns. Tier counts are 12 Calamity, 34 Menace, 99 Wanted and 20 Unknown. Three anonymous descriptions are retained and flagged. The bounty distribution is highly unequal (Gini 0.75); the top five characters hold 42.2% of known bounty and the top 10% hold 64.2%. Among name-based features, Has_D_Clan has the strongest Spearman correlation with log bounty at 0.28, indicating only a weak relationship. The median log10 bounty is 9.18 for the five known D. clan members and 7.95 for the rest; the small group means this is descriptive, not proof of an effect.

4. Result and conclusion

Feature engineering adds scaling, tier, text, grouping and target variables for later analysis. The EDA indicates strong inequality and weak predictive signal in name text alone.

5. Modelling caution: target leakage

Tier, percentile and High_Bounty are calculated from bounty itself. Exclude these features from predictor sets when the modelling objective is to predict bounty from independent information; otherwise the model can indirectly receive the answer.

Appendix — Original EDA Report, Page 1

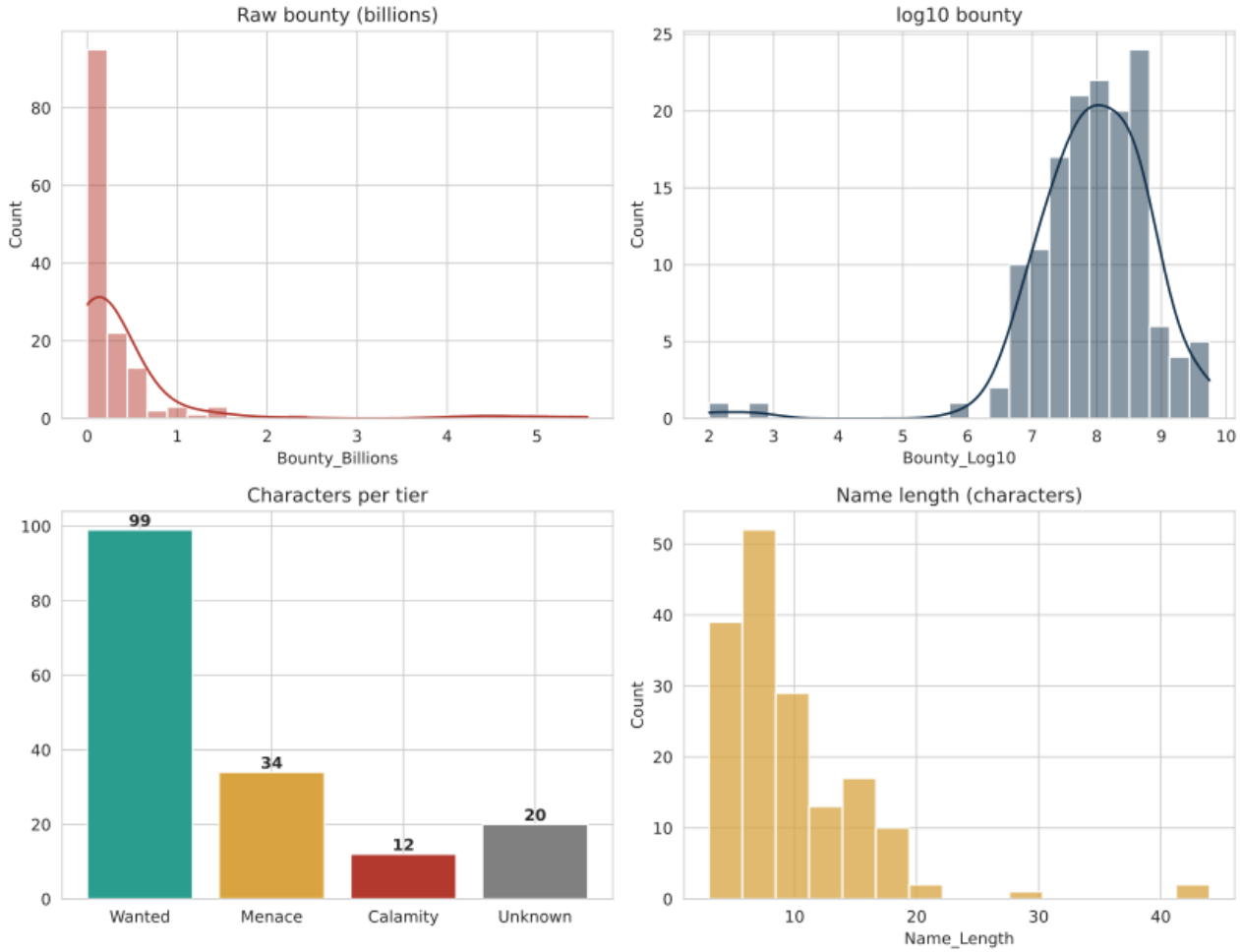
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SKILL 3 - FEATURE ENGINEERING AND EDA
One Piece Bounty Dataset

Rows: 165 | Known bounties: 145 | Unknown: 20
Columns: 3 raw -> 23 engineered

Tier rules
  Calamity : bounty >= 1,000,000,000
  Menace   : 300,000,000 to 999,999,999
  Wanted   : below 300,000,000

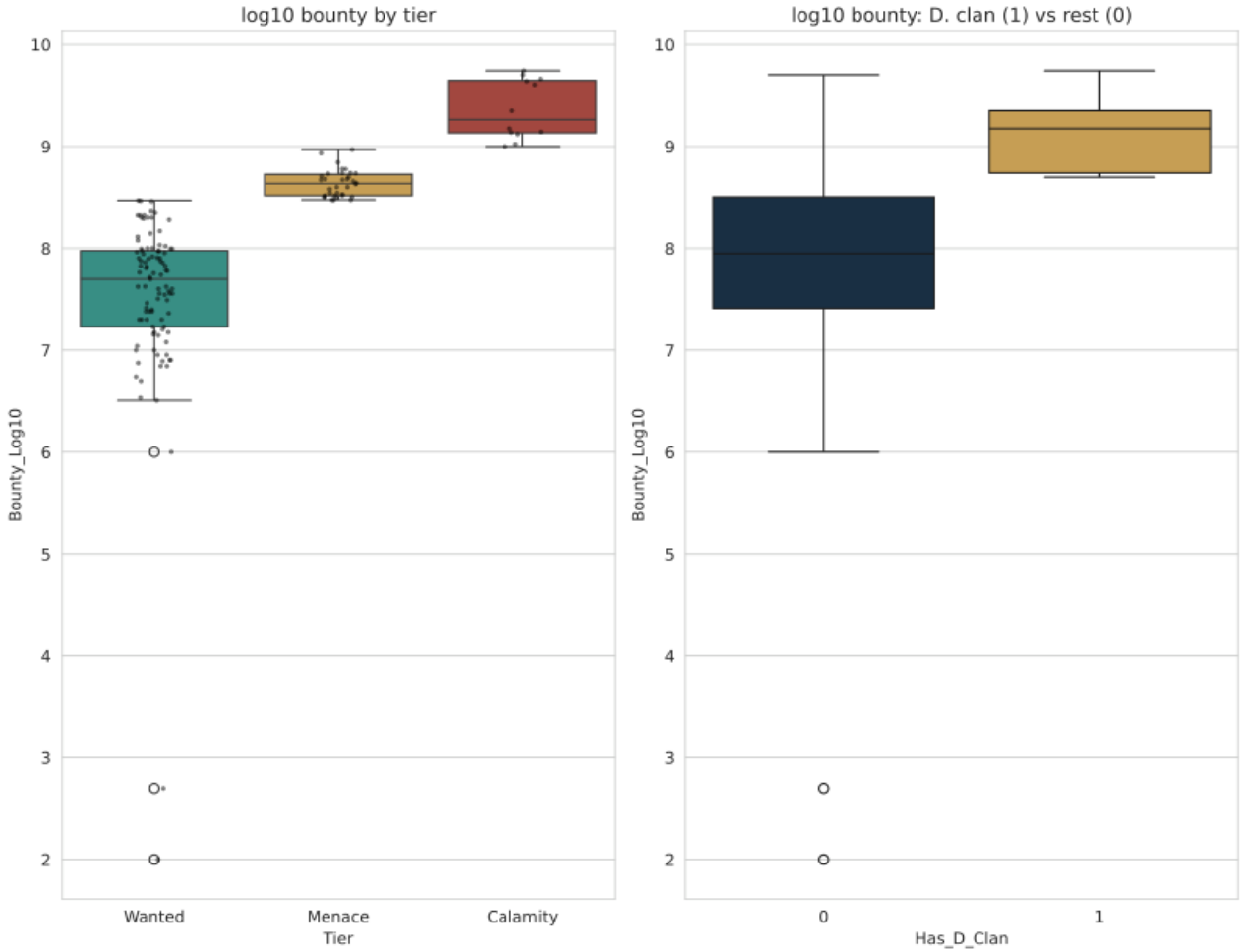
Feature groups
  Scaling   : Billions, Log10, Percentile, Z-score, Min-Max
  Binning   : Tier, Tier_Code, one-hot Tier columns
  Text      : Name_Length, Word_Count, Has_D_Clan, Has_Parentheses, Is_Anonymous
  Grouping  : Name_Group, Group_Size (frequency encoding), Is_Charlotte
  Target    : High_Bounty (log bounty >= median)
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Appendix — Original EDA Report, Page 2

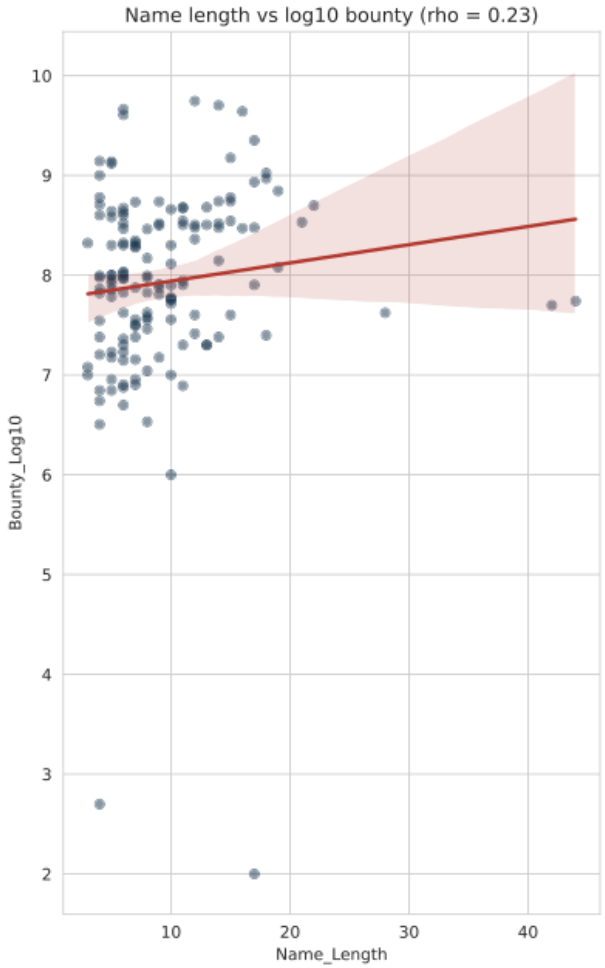


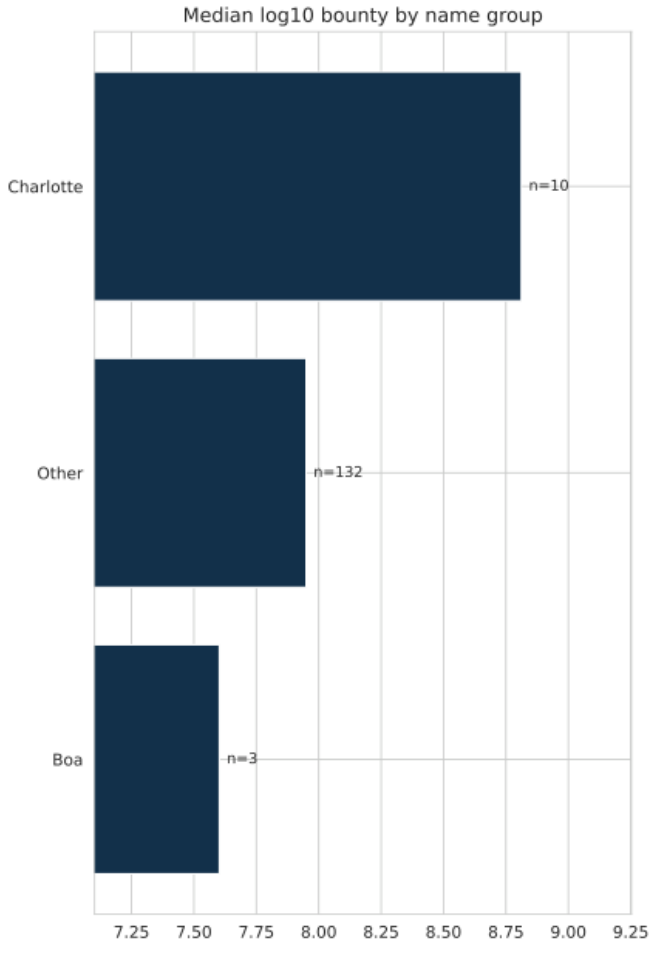
The distribution charts compare raw bounty values, log10 bounty, tier counts and name length.

Appendix — Original EDA Report, Page 3

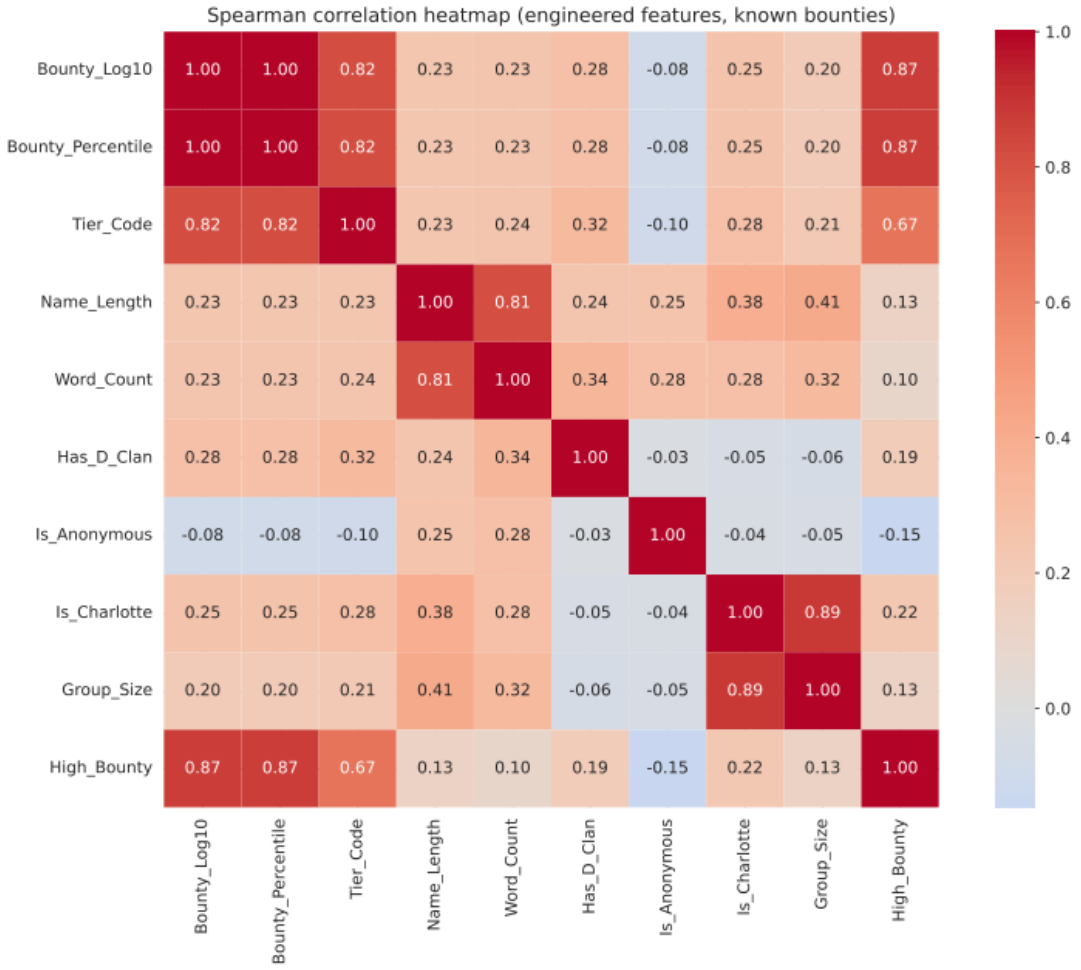


Appendix — Original EDA Report, Page 4



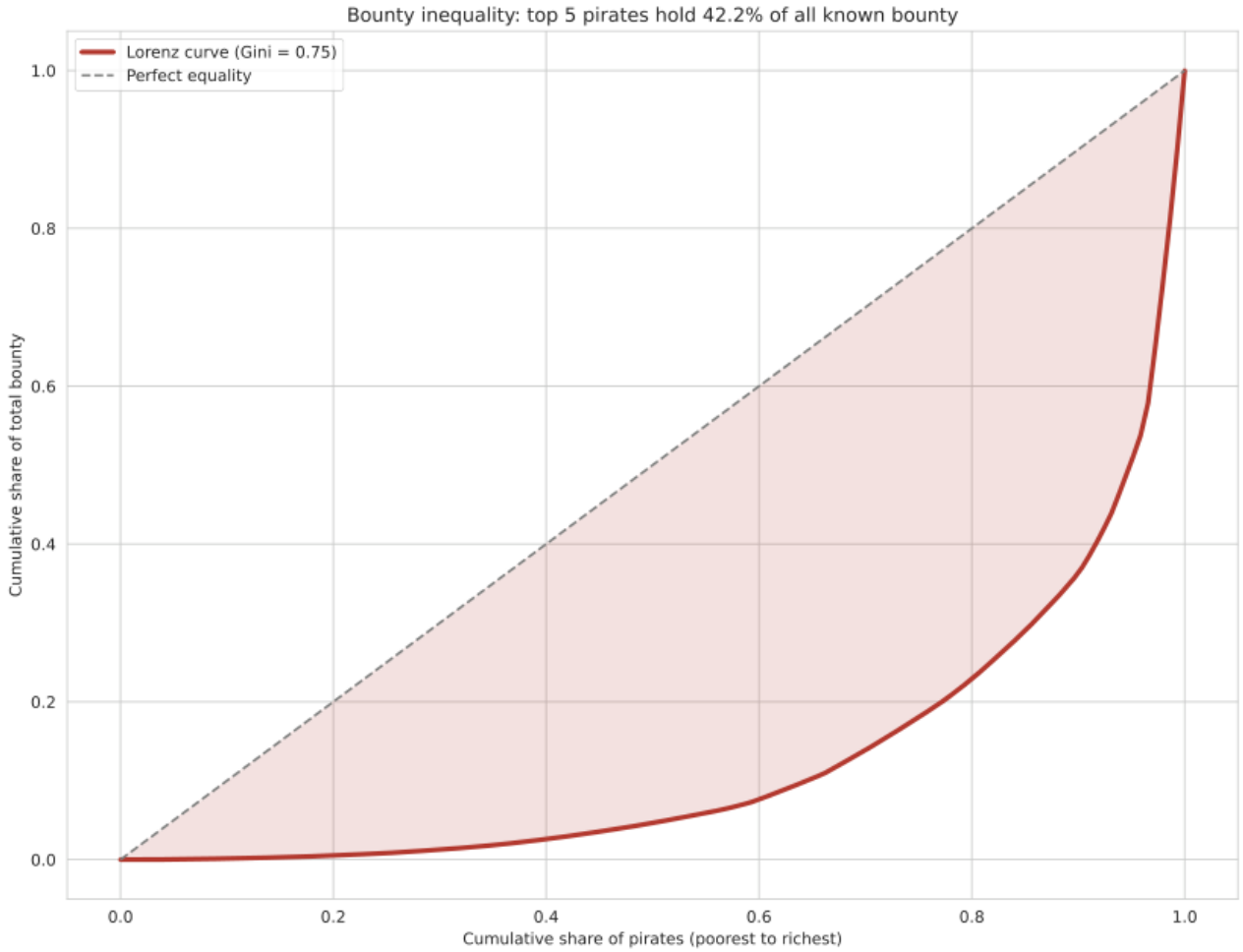


Appendix — Original EDA Report, Page 5



The correlation heatmap shows that name-based features have weaker relationships with log bounty than bounty-derived features.

Appendix — Original EDA Report, Page 6



The Lorenz curve summarizes bounty inequality; the reported Gini coefficient is 0.75.

Appendix — Original EDA Report, Page 7

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FEATURE DICTIONARY

Bounty_Billions      float64  nulls=20
Bounty_Log10         float64  nulls=20
Bounty_Percentile    float64  nulls=20
Bounty_Log_Zscore    float64  nulls=20
Bounty_Log_MinMax    float64  nulls=20
Tier                 object   nulls=0
Tier_Code            float64  nulls=20
Tier_Calamity        int64    nulls=0
Tier_Menace          int64    nulls=0
Tier_Unknown         int64    nulls=0
Tier_Wanted          int64    nulls=0
Is_Anonymous         int64    nulls=0
Name_Length          int64    nulls=0
Word_Count           int64    nulls=0
Has_D_Clan           int64    nulls=0
Has_Parentheses      int64    nulls=0
Name_Group           object   nulls=0
Group_Size           int64    nulls=0
Is_Charlotte         int64    nulls=0
High_Bounty          float64  nulls=20
```